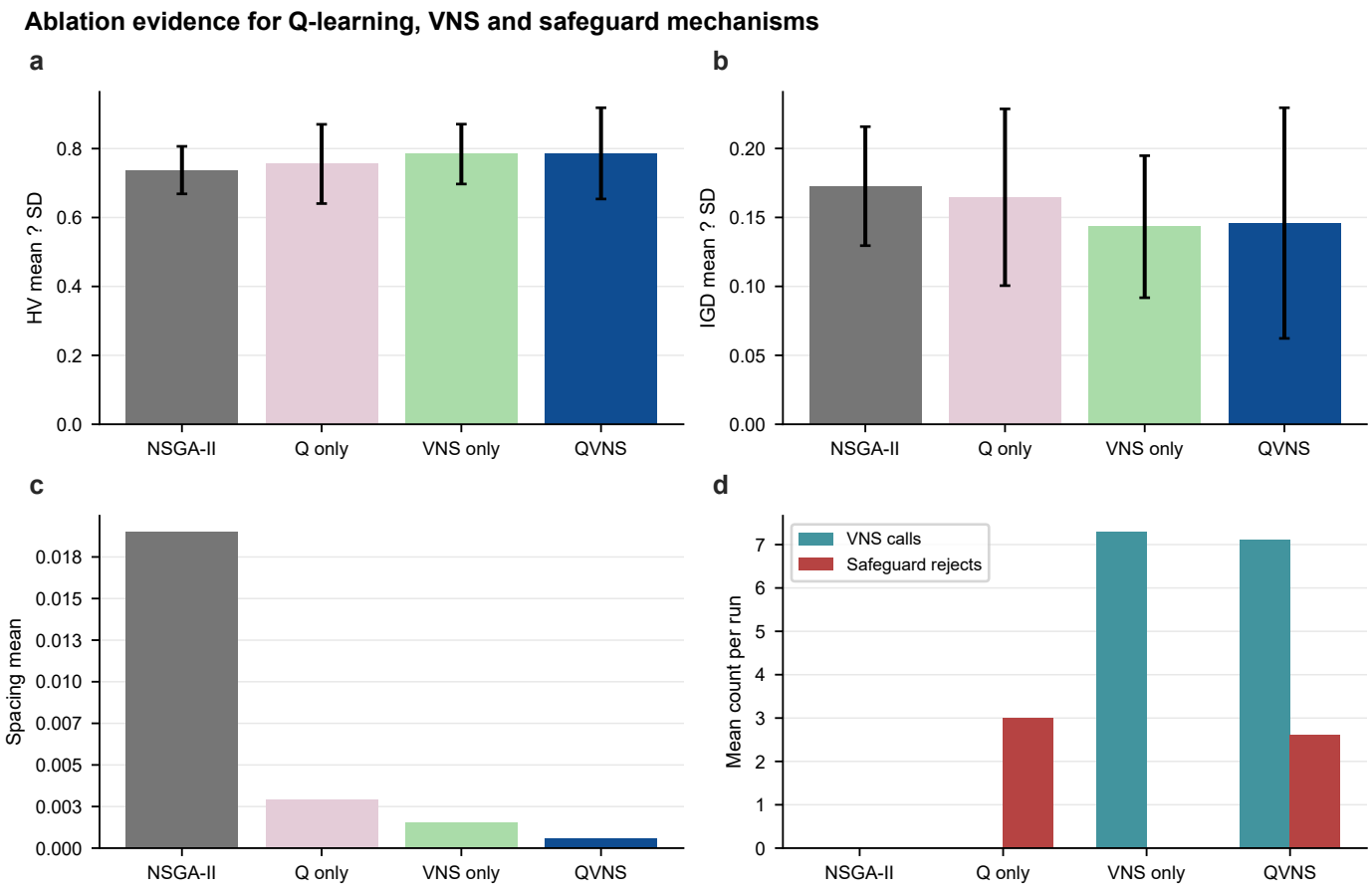
Ablation evidence for Q-learning, VNS and safeguard mechanisms
a
HV mean ? SD
0.0
0.2
0.4
0.6
0.8
NSGA-II
Q only
VNS only
QVNS
b
IGD mean ? SD
0.00
0.05
0.10
0.15
0.20
NSGA-II
Q only
VNS only
QVNS
c
Spacing mean
0.000
0.003
0.005
0.007
0.010
0.013
0.015
0.018
NSGA-II
Q only
VNS only
QVNS
d
Mean count per run
VNS calls
Safeguard rejects
0
1
2
3
4
5
6
7
NSGA-II
Q only
VNS only
QVNS